Title: Rethinking Crypto-inventory Discovery in Enterprises: A Isogeny-based Protocols Perspective

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Abstract:

Crypto-inventory discovery in enterprises remains a critical challenge for post-quantum cryptography, particularly in banking infrastructure. This paper proposes a framework built on isogeny-based protocols that explicitly accounts for harvest-now-decrypt-later risk. We evaluate the framework on a simulated 100 km fiber QKD channel and compare it against 7 baseline methods. Experimental results show that the proposed approach improves decapsulation time by 13.3% while keeping signature size within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of post-quantum cryptography has changed how researchers and practitioners approach crypto-inventory discovery in enterprises, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Over the past decade, isogeny-based protocols have attracted sustained attention as a promising route toward solving crypto-inventory discovery in enterprises in realistic settings such as banking infrastructure. Despite considerable progress, existing solutions for crypto-inventory discovery in enterprises tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. This paper addresses these gaps by studying how isogeny-based protocols interact with lattice-based key encapsulation when applied to crypto-inventory discovery in enterprises.

Our work is motivated by the growing demand for banking infrastructure, where reliability and transparency requirements are especially strict. We make three contributions: a unified formulation of crypto-inventory discovery in enterprises, a practical design built on isogeny-based protocols, and an empirical evaluation on a simulated 100 km fiber QKD channel. Understanding harvest-now-decrypt-later risk is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Alvarez et al. (2014) introduced an early framework based on isogeny-based protocols, reporting a 5.8% improvement in secret key rate over previous baselines. Subsequent work by Rossi et al. (2019) extended this line of research to banking infrastructure, although the evaluation was restricted to small-scale settings. Researchers working on post-quantum cryptography have explored lattice-based key encapsulation as an alternative, with OConnor et al. (2021) noting that the approach scales poorly beyond 1200 components. Mukherjee et al. (2019) analysed harvest-now-decrypt-later risk and showed that purely heuristic solutions cannot guarantee consistent signature size. More recently, Okafor et al. (2011) combined isogeny-based protocols with lattice-based key encapsulation, achieving a 20.5% reduction in overhead on a simulated 100 km fiber QKD channel.

Several studies have criticised the reliance on synthetic workloads, arguing that results on a simulated 100 km fiber QKD channel may not generalise to banking infrastructure. In contrast to these efforts, our work treats crypto-inventory discovery in enterprises and harvest-now-decrypt-later risk jointly rather than as independent

concerns. Industry reports corroborate these academic findings, pointing to crypto-inventory discovery in enterprises as a persistent obstacle for banking infrastructure. Despite this body of literature, no prior study has systematically compared isogeny-based protocols and lattice-based key encapsulation under identical experimental conditions.

3. Methodology

The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on isogeny-based protocols, and a validation layer that incorporates lattice-based key encapsulation. All experiments were conducted on a simulated 100 km fiber QKD channel, which provides 500 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 20 times with independent random seeds, and we report mean values together with 95% confidence intervals. Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data.

The core routine applies isogeny-based protocols to reduce the effect of harvest-now-decrypt-later risk, followed by a calibration step that stabilises logical qubit count. Our implementation is written in Python and C++, and was executed on a workstation with 64 GB of memory and a single mid-range accelerator. We compare against 7 baseline methods, including a conventional approach built on lattice-based key encapsulation and two recent variants reported in the literature. The proposed design improves decapsulation time by 13.3% relative to the strongest baseline and reduces decapsulation time by 28.0%, a statistically significant difference ($p < 0.05$).

Ablation experiments show that removing the lattice-based key encapsulation component reduces decapsulation time by approximately 12.5%, confirming its contribution. A sensitivity analysis over 16 parameter settings shows that performance remains stable, with variation in signature size below 1.5%. Threats to validity include the limited diversity of a simulated 100 km fiber QKD channel and the possibility that deployment in banking infrastructure introduces conditions not captured in our setup.

4. Conclusion

This paper investigated crypto-inventory discovery in enterprises through the lens of isogeny-based protocols, and demonstrated that a carefully designed pipeline can improve decapsulation time by 13.3% on a simulated 100 km fiber QKD channel. The findings suggest that post-quantum cryptography will continue to benefit from closer integration of isogeny-based protocols and lattice-based key encapsulation, particularly in banking infrastructure. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of harvest-now-decrypt-later risk. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for crypto-inventory discovery in enterprises.